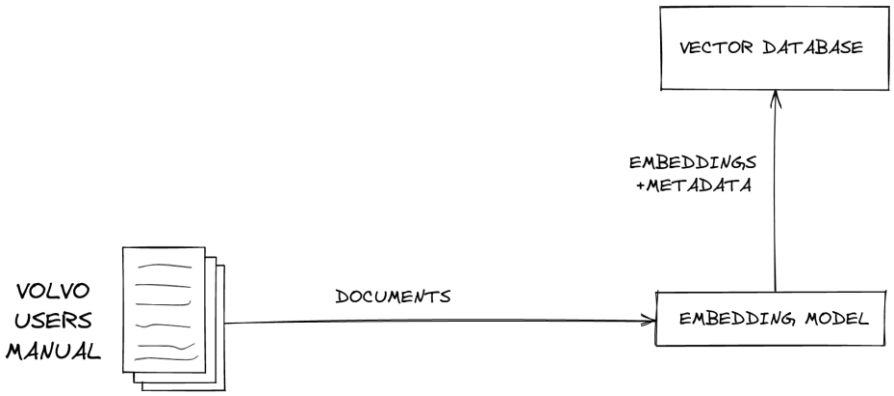
Week 2: Retrieval Augmented Generation

- **What is Retrieval Augmented Generation?**
  - Retrieval Augmented Generation, or RAG, is a sophisticated architecture that improves the performance of LLMs, i.e. large language models (e.g. ChatGPT or Gemini). The gist is that it retrieves relevant information from external knowledge, and provides that information to the model at question time in real-time.
- **Why do we need RAG?**
  - Considering their capabilities, LLMs have shortcomings as well.
  - While LLMs can produce high-quality answers, RAG enables LLMs to produce high-quality answers more accurately, more contemporaneously, and more tailored to users by providing LLMs with access to new and unique data exactly when it is needed.
- **Limitations of LLMs (without using RAG):**
  - Outdated knowledge & information: LLMs have fixed datasets that inhibit them from answering questions about recent events or developments.
  - Lack of company-specific information: LLMs do not have any access to your internal or proprietary data.
  - Hallucinations: LLMs can sometimes confidently produce incorrect or fabricated information.
  - Opaque reasoning: It's difficult to recognize the provenance of their answers.
  - Costly fundamentally retraining: Retraining LLMs with new data & information is expensive and time-consuming.
- **How does RAG actually work?**
  - You ask a question.
  - A retrieval system queries a knowledge base or vector database to identify relevant documents.
  - The context is passed on to the LLM, which it uses to generate an answer that has been produced more reliably and with unadulterated context.



RAG is the most cost-effective, easy-to-implement, and lowest-risk path to higher performance for GenAI applications.

RAG vs Fine-Tuning

- ***RAG - Retrieval Augmented Generation***
  Think of it like the smartest first bencher of your class.
  Instead of depending on his memory and the things he has studied, he looks into his study materials, ppt, and notes to answer tricky questions, just like RAG - it retrieves info from external sources like dbs or docs and generates the accurate response.
- ***What problem does it solve?***
  Now imagine you ask a difficult question to your first bencher friend. He panics, because he knows the topic but doesnt remember the exact answer. But like RAG, he quickly looks into his notes then **retrieves the information** and gives you the correct answer**(generation of response).**
- So now ***ChatGPT*** is like a COVID-era kid who studied hard till 2023. After that no updates. Ask something recent about him, he might go blank. But RAG, he is always updated.
- ***Why RAG over fine-tuning?***
  - Fine-tuning is like pulling an all-nighter before the exam, stuffing all the knowledge into your brain, stressful and with no sleep hoping you remember something in the exam.
  - While RAG is like keeping up with the syllabus, no backlogs, just chill with full sleep during semester and scoring 9.5 CGPA.